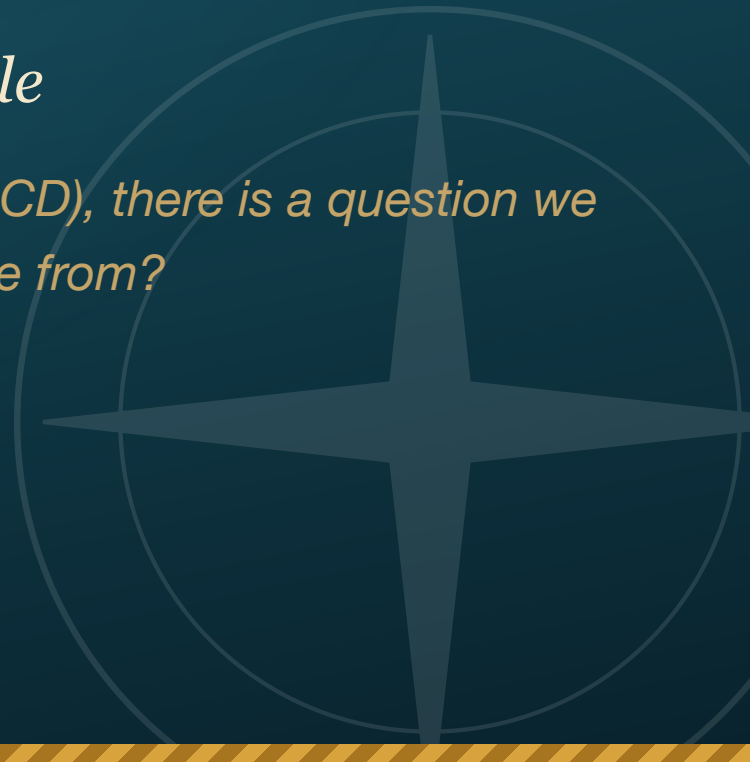
DAY 3 · OPENING DEMO · 30 MINUTES

Gitea Actions

In-cluster CI, GitHub-Actions-compatible

Before we template the manifests (Helm) and before we deploy them (ArgoCD), there is a question we have been quietly skipping: where does the image come from?



Where does the image come from?

- For two days, *we* built the images — locally, with `docker build`, then `docker push` to Harbor.
- That isn't a workflow. That's the instructor cheating off-stage.
- After lunch you'll meet **ArgoCD**, which pulls images from Harbor and ships them to the cluster. But ArgoCD assumes the image is **already there**.
- This morning we fix that: a `git push` automatically produces a tagged image in Harbor, with no human in the middle.

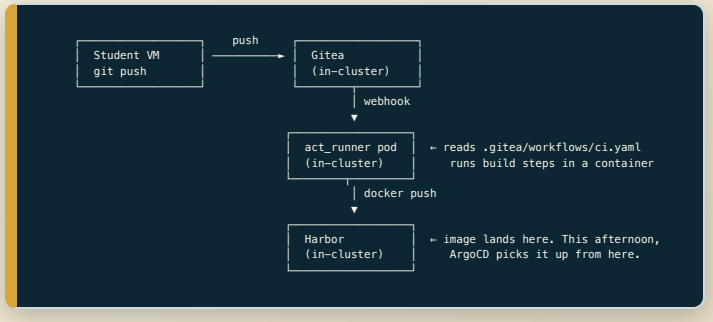
> *Today we build the build half. After lunch ArgoCD finishes the loop.*

The committed shape (and what we actually built)

- The original course outline named **GitHub Actions**: workflows, jobs, secrets, build-and-push.
- We will **not** use GitHub Actions in the cluster. The room has no public internet path and the cluster shouldn't depend on github.com.
- Instead: **Gitea Actions** — the same workflow YAML, executed by an in-cluster runner against our in-cluster Gitea.
- *Same mental model. Different home.*

Concept	GitHub Actions	Gitea Actions
Workflow file	`.github/workflows/ci.yaml`	`.gitea/workflows/ci.yaml`
Runner	GitHub-hosted or self-hosted	`act_runner` pod in our cluster
Image cache	github.com Actions cache	local registry / runner volume
Marketplace	`uses: docker/build-push-action@v5`	**same** — Gitea Actions is `act`-compatible

What's actually running



Every box on this diagram is a pod in your cluster.

The workflow file

```yaml
# .gitea/workflows/ci.yaml
name: build-and-push
on:
  push: { branches: [main] }

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Log in to Harbor
        uses: docker/login-action@v3
        with:
          registry: harbor.${{ vars.LAB_DOMAIN }}
          username: ${{ secrets.HARBOR_ROBOT_USER }}
          password: ${{ secrets.HARBOR_ROBOT_SECRET }}

      - name: Build and push
        uses: docker/build-push-action@v6
        with:
          push: true
          tags: harbor.${{ vars.LAB_DOMAIN }}/raft-fleet/raft:${{ github.sha }}
```

That's it. Same syntax a working Actions engineer writes on day one.

Live demo — push to image

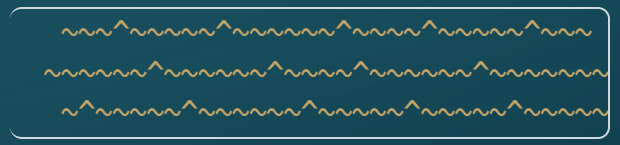
We're going to push a one-character change to a repo and watch it produce a tagged image in Harbor, with no human in the middle.

1. **Edit** a string in the `raft` repo's `index.html`.
2. **Commit + push** to Gitea.
3. Watch the **Actions** tab in Gitea — runner picks up the job within 5 seconds.
4. Watch the build logs stream in real time.
5. Watch **Harbor** — a new tag appears under `raft-fleet/raft` when push completes.

From `git push` to tagged image in under 60 seconds. The image is now sitting in Harbor, waiting for something to deploy it.

WHAT'S NEXT ON THE SHELF

Beyond Gitea Actions



When Gitea Actions stops being enough

- **Gitea Actions** is the right answer for: classroom labs, small teams, GitHub-compatible workflows.
- It's the *wrong* answer when you need:
  - DAG topologies (fan-out / fan-in across many steps with conditional dependencies)
  - **CRD-native** pipelines that other Kubernetes tools can introspect
  - First-class CNCF graduation status for your platform's compliance review

Two CNCF-graduated projects exist for exactly that.

Argo Workflows — the Argo family answer

- **CNCF graduated.** Same project family as ArgoCD.
- Pipelines are CRDs — `Workflow`, `WorkflowTemplate`, `CronWorkflow`.
- Designed around **DAGs** — every step is a container, dependencies are explicit.
- Argo **Events** (sibling project) gives you webhook / git / S3 / kafka triggers.
- Most natural fit if you're already standing up ArgoCD — same UI conventions, same RBAC story.

```yaml
apiVersion: argoproj.io/v1alpha1
kind: Workflow
spec:
  templates:
    - name: build
      container: { image: gcr.io/kaniko-project/executor, args: [ ... ] }
```

Tekton — the industry-standard answer

- **CNCF graduated.** The reference Kubernetes-native CI engine.
- Pipelines are CRDs — `Task`, `Pipeline`, `PipelineRun`, `TaskRun`.
- 1:1 conceptual map to GitHub Actions: workflow → pipeline, job → task, step → step.
- **Tekton Hub** is a large catalog of reusable Tasks (build, scan, sign, deploy).
- The one you'll see in industry CI/CD platforms (OpenShift Pipelines is Tekton, Jenkins X uses Tekton, etc.).

```yaml
apiVersion: tekton.dev/v1
kind: Pipeline
spec:
  tasks:
    - name: build-image
      taskRef: { name: buildah, kind: ClusterTask }
```

How to choose, in one sentence

If you want…	Reach for…
GitHub Actions semantics, in-cluster, today	**Gitea Actions**
DAGs in the same UI as your ArgoCD	**Argo Workflows**
CNCF reference platform, big task catalog	**Tekton**

All three run on the same cluster you already have. None of them is wrong.

Now: meet Hazel.

You've seen the build half. The template + deploy halves are the rest of today.